MODULE 1 OF 5

Introduction & Clinical Rationale

~22 min read Running Gait Analysis for Injury Prevention & Treatment

1.1 Welcome & Course Overview

This course is designed for physiotherapists who already have a strong working knowledge of anatomy, biomechanics, and general musculoskeletal assessment, but want a structured, evidence-based framework specifically for running gait analysis. Rather than re-teaching anatomy from first principles, each module applies your existing clinical knowledge to the specific demands, movement patterns, and injury profiles seen in runners.

Across the course you will move from the general (why gait analysis matters and what it can and can't do) through to the specific (how to actually run the assessment, interpret what you see, treat accordingly, and build a return-to-run programme). This module lays the groundwork: the scale of the running-injury problem, the honest evidence base behind gait analysis and gait retraining, and where this fits within a broader clinical assessment.

How This Course Works

- Five modules, each building on the last — starting with rationale and evidence, then methodology, interpretation, treatment, and return-to-run programming.
- Content assumes physiotherapy-level anatomy and clinical reasoning; running-specific context is explained in full.
- Each module includes practical frameworks, tables, and case-relevant detail you can apply directly in a clinic setting.
- Modules 3 and 4 are structured around the link between gait faults and specific pathologies, reflecting how runners actually present.

1.2 The Scale of the Problem

Running is one of the most accessible and widely practised forms of exercise, but running-related injuries (RRIs) are common enough that most physiotherapists will see them regularly regardless of whether they consider themselves a "running specialist." Reported incidence figures vary considerably depending on how a study defines injury and which running population it studies, but the pattern is consistent: this is a high-volume clinical problem, not a niche one.

A widely cited meta-analysis found a weighted injury incidence of 17.8 injuries per 1000 hours of running in novice runners, compared with 7.7 per 1000 hours in recreational runners — meaning novice runners face substantially higher injury risk per hour run than more experienced ones, likely reflecting a mismatch between tissue capacity and training exposure early on.

Pathology	Approximate Share of RRIs
Patellofemoral pain syndrome	Consistently among the most common presentations across incidence and prevalence studies
Medial tibial stress syndrome	One of the leading lower-leg overuse presentations
Achilles tendinopathy	Consistently among the top five most common RRIs
Plantar fasciitis	Consistently among the top five most common RRIs
Iliotibial band syndrome	Consistently among the top five most common RRIs

Sources: Francis P, Whatman C, Sheerin K, Hume P, Johnson MI. The Proportion of Lower Limb Running Injuries by Gender, Anatomical Location and Specific Pathology: A Systematic Review. J Sports Sci Med. 2019;18:21–31. Videbæk S, Bueno AM, Nielsen RO, Rasmussen S. Incidence of Running-Related Injuries Per 1000 h of Running in Different Types of Runners: A Systematic Review and Meta-Analysis. Sports Med. 2015;45:1017–1026.

OFTEN-CITED FRAMING

You'll frequently hear running-injury rates summarised as "roughly 80% of runners get injured every year" — a widely used rule-of-thumb in running-injury teaching and media. In reality, reported annual incidence varies enormously by study (commonly cited ranges run anywhere from around 20% up to 70–90%+) depending on how "injury" is defined and which running population is sampled. The honest takeaway is the same either way: this is a very common problem, not a precise, agreed-upon number — and that ambiguity is itself worth naming explicitly to a sceptical colleague or CPD audience rather than quoting a single figure as settled fact.

A systematic review focused specifically on running-related musculoskeletal injuries found broadly consistent prevalence ranges across studies: medial tibial stress syndrome around 9.5%, Achilles tendinopathy roughly 6.2–9.5%, and plantar fasciitis 5.2–17.5% among runners generally — with marathon-specific samples showing Achilles tendinopathy as high as 2.0–18.5% and patellofemoral pain syndrome 7.4–15.6%. These sit alongside, rather than replace, the incidence figures above: prevalence (how many runners have the condition at a point in time) and incidence (how many per hours run) are different measures, and both are reported inconsistently enough in the literature that ranges, not single numbers, are the honest way to report them.

Source: Lopes AD, Hespanhol Júnior LC, Yeung SS, Costa LOP. What are the Main Running-Related Musculoskeletal Injuries? A Systematic Review. Sports Med. 2012;42(10):891–905.



These five pathologies will reappear throughout Modules 3 and 4, since they cover the large majority of what you'll actually see walk through the door. Knowing which injuries dominate the clinical picture is itself useful context before you even pick up a camera — it tells you what a gait analysis needs to be sensitive to.

1.3 Why Look at Gait at All?

Running-related injury is widely accepted to be multifactorial — training load, tissue capacity, strength, footwear, surface, sleep, and psychosocial factors all contribute alongside biomechanics. Gait analysis is not a replacement for a full clinical assessment; it is one lens within it. The rationale for including it is threefold:

- **It's a modifiable factor.** Unlike a runner's anatomy or training history to date, gait pattern is something a clinician can realistically intervene on within a treatment plan.
- **It gives you a shared, visual reference point with the patient.** Showing a runner their own footage tends to make abstract biomechanical concepts concrete and improves buy-in for a retraining or exercise programme.

- **It can reveal compensations that a static assessment misses.** Some deficits (hip control, cadence, vertical oscillation) only become apparent under the repeated, fatiguing load of actual running.

That said, the honest starting position for this course is that gait analysis is a useful adjunct, not a diagnostic gold standard — which is exactly why Modules 1.4 and 1.5 below spend real time on what the evidence does and doesn't support.

1.4 What the Evidence Shows

The most direct evidence on whether changing gait actually helps comes from studies of **gait retraining** — deliberately modifying an aspect of running technique, usually with real-time feedback, and measuring what happens to biomechanics, pain, and function.

A 2022 systematic review with meta-analysis of 19 randomised controlled trials found moderate-certainty evidence that step-rate (cadence) retraining reliably increases cadence and reduces certain loading-related variables. The same review found only low-to-moderate certainty evidence that switching to a non-rearfoot strike pattern, or multiparameter interventions more broadly, changed running economy or performance one way or the other — in other words, cadence retraining has a relatively solid evidence base, while foot-strike modification does not, at least for effects on performance.

A more recent umbrella review focused specifically on patellofemoral pain in runners synthesised results across several systematic and narrative reviews and found the most consistent supporting evidence for **cadence increases of roughly 5–10%** and **soft-landing cues to reduce vertical impact**. Findings for foot-strike modification and for injury prevention (as opposed to short-term pain and function) specifically were considerably less consistent.

Gait Retraining Strategy	Strength of Supporting Evidence
Cadence increase (~5–10%)	CONSISTENT (SHORT-TERM) Most consistent evidence for reducing loading variables and short-term pain/function gains
Soft-landing / reduced vertical impact cues	CONSISTENT (SHORT-TERM) Consistent short-term improvements in pain and loading-related variables
Foot-strike pattern modification	MIXED Mixed — changes kinematics, but effect on performance, pain, and injury risk is inconsistent
Multiparameter interventions	LOW CERTAINTY Feasible, but current trials give only low-certainty evidence on clinical outcomes

Sources: Doyle E, Doyle TLA, Bonacci J, Fuller JT. The Effectiveness of Gait Retraining on Running Kinematics, Kinetics, Performance, Pain, and Injury in Distance Runners: A Systematic Review With Meta-analysis. J Orthop Sports Phys Ther. 2022;52:192–A5. Alexander JLN, Culvenor AG, Johnston RRT, Ezzat AM, Barton CJ. Gait Retraining for Patellofemoral Pain in Runners: An Umbrella Review of Clinical and Biomechanical Evidence. Sports Med Open.

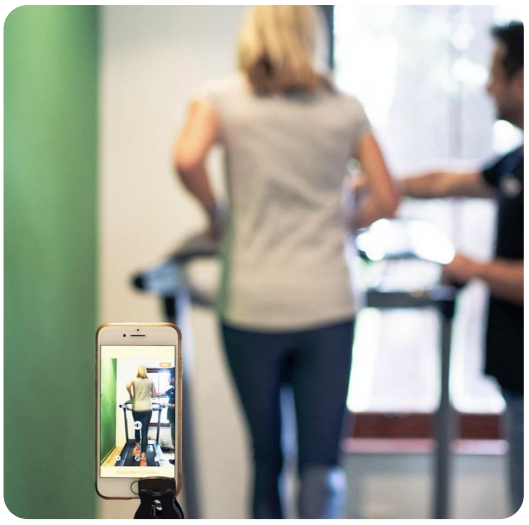
1.5 The Real Limits of This Evidence

It's tempting, once you've seen a gait analysis "work" for one patient, to treat the method as more validated than it is. Two limitations are worth holding onto throughout this course:

Visual gait analysis itself has only moderate reliability

A frequently cited reliability study asked ten raters — experienced, inexperienced, and expert — to independently score videotaped gait patterns twice each. Inter-rater reliability among experienced raters was only moderate (ICC ≈ 0.42), comparable to inexperienced raters, and even expert raters only reached moderate-to-good agreement

(ICC ≈ 0.54). In other words, two competent clinicians watching the same run may not consistently agree on what they're seeing — which is exactly why Module 2 emphasises a structured, repeatable method rather than free-form observation.



A standard phone-and-tripod gait analysis set-up in a clinic setting.

Standardised 2D video-based analysis (fixed camera position, defined landmarks, consistent frame rate) performs meaningfully better than unstructured visual observation, and has been shown to correlate reasonably well with 3D motion capture for several key variables — which is the whole rationale for the structured video protocol taught in Module 2, rather than an unstructured "watch them run and see what looks off" approach.

Gait retraining evidence is still an emerging field

Even where gait retraining shows benefit, most trials are small, short-term, and focused on a handful of conditions (patellofemoral pain in particular). Evidence for gait retraining as a genuine long-term injury-prevention strategy — as opposed to a short-term pain and loading intervention — remains limited. This doesn't mean gait retraining isn't worth doing; it means it should be positioned to patients and documented in notes as one evidence-informed component of a broader plan, not an infallible fix.

> **CLINICAL NOTE**
>
> Because inter-rater reliability of unstructured gait observation is only moderate even among experienced clinicians, your confidence in any single gait analysis finding should scale with how standardised your method was — camera position, view angle, treadmill vs overground, and defined criteria for what you're scoring all matter. Module 2 sets out that structured method in full.
>
> *Reference: Brunnekreef JJ, van Uden CJT, van Moorsel S, Kooloos JGM. Reliability of videotaped observational gait analysis in patients with orthopedic impairments. BMC Musculoskelet Disord. 2005;6:17.*

1.6 Where Gait Analysis Fits Clinically

Given the above, gait analysis is best understood as one input into a broader clinical reasoning process, not a standalone diagnostic tool. A defensible clinical workflow places it alongside, rather than instead of:

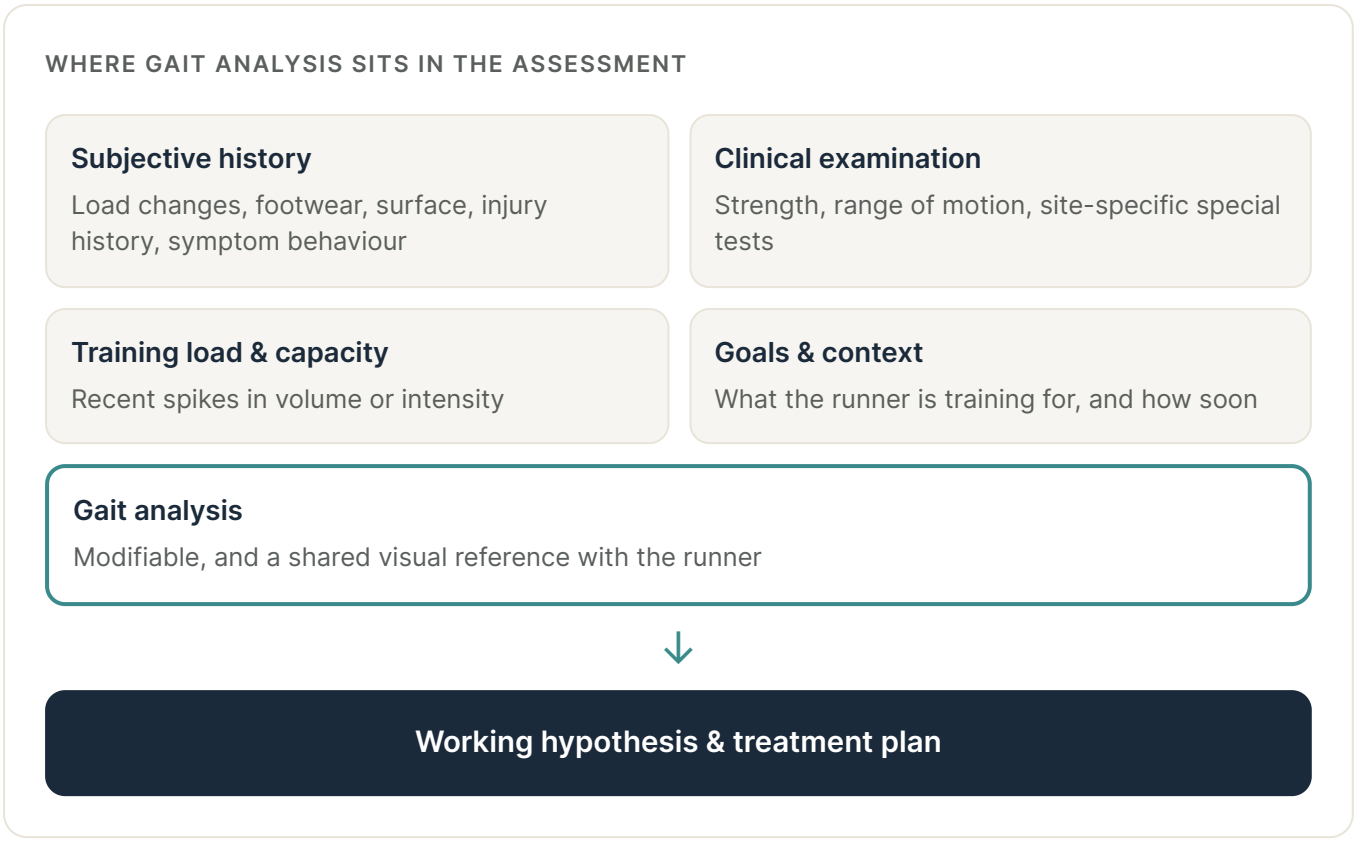
- **Subjective history** — training load and its recent change, footwear, surface, prior injury history, and symptom behaviour.
- **Clinical examination** — strength, range of motion, and site-specific special tests relevant to the presenting complaint.
- **Training load and capacity** — since a sudden spike in volume or intensity is one of the most consistently implicated risk factors across the running-injury literature, independent of gait pattern.
- **The patient's own goals and context** — a recreational runner returning from a first-ever injury and an athlete chasing a qualifying time need different framing even with identical gait findings.



From subjective history, through clinical examination, to on-track review — gait analysis sits within this broader assessment, not apart from it.

Gait analysis earns its place in this workflow because it is genuinely modifiable and gives you and the runner a shared reference point — not because it is more objective or more predictive than the rest of the assessment. Module 3 will return to this integration point

directly when it comes to linking specific gait observations to specific pathologies.



Gait analysis is one input alongside four others; no single input decides the plan on its own.

1.7 What This Course Will Cover

With that context in place, the remaining four modules build a complete, repeatable clinical pathway:

- **Module 2 — Running Gait Analysis Methodology:** a structured, standardised video-based method (camera setup, views, and what to look for) designed to reduce the reliability problems discussed above.
- **Module 3 — Interpreting Gait & Linking to Pathology:** reading specific gait patterns and mapping them to the injuries covered in section 1.2.
- **Module 4 — Injury-Specific Treatment Pathways:** per-injury clinical reasoning and how to sequence treatment against your assessment findings.
- **Module 5 — Running Re-Training & Return-to-Run Programming:** cueing, drills, load management, and a structured return-to-run progression, built on the cadence and soft-landing evidence discussed above.

What this course will *not* do is present gait analysis as a stand-alone diagnostic test, or claim stronger evidence for any single gait-retraining strategy than the literature currently supports. The goal throughout is a clinically useful, honestly evidenced framework you can defend to a sceptical colleague — not a set of confident-sounding rules that outpace what's actually been shown.

Summary

Running injuries are common, multifactorial, and dominated by a relatively small set of recurring pathologies. Gait analysis is a genuinely useful, modifiable piece of the clinical picture — but visual gait observation is only moderately reliable without a structured method, and gait retraining evidence, while real, is strongest for cadence and soft-landing strategies specifically rather than as a blanket claim. With that grounding in place, Module 2 moves into the structured methodology itself.